Abu Dhabi City Campus Daily/Weekly Planner

Week 10: 2nd – 6th November 2026

Subject: Mathematics Diploma Programme **Unit Title: Geometry and Trigonometry / Internal Assessment** **Grade: 11**

Lessons	Learning targets	Assessment tools and strategies Formative/ Summative/ Differentiation	Resources and links for Parents
Lesson 1: Perpendicular Bisectors	• I can construct and determine the equation of a perpendicular bisector by finding a segment's midpoint and using the negative reciprocal of its gradient.	2.1 Clarify vocabulary and concepts: midpoint, gradient, perpendicular bisector, negative reciprocal.	Lesson notes and practice on Teams/OneNote. Perpendicular bisector [IB Maths AI SL/HL] Intro to Voronoi diagrams [IB Maths AI SL/HL] Voronoi diagrams - adding sites, nearest neighbour [IB Maths AI SL/HL] Voronoi diagrams - toxic waste dump [IB Maths AI SL/HL]
Lesson 2: Introduction to Voronoi Diagrams	• I can explain how perpendicular bisectors partition a plane into regions and construct a Voronoi diagram from given sites.	5.3 Build fluency with graduated scaffolds: begin with two sites and one perpendicular bisector, then add further sites and boundaries.	
Lesson 3: Applications of Voronoi Diagrams	• I can apply Voronoi diagrams to real-world location problems and justify which site is closest to a given point or region.	8.2 Vary challenge level: support students with a prepared diagram; extend students to evaluate whether a Voronoi model is realistic when roads, population or barriers matter.	
Lesson 4: Introduction to the Internal Assessment (IA)	• I can explain the purpose and structure of the IB Mathematics Internal Assessment (IA) and identify the main criteria used to assess.	9.3 Develop self-assessment and peer feedback: students use a checklist to evaluate whether their draft question is focused, mathematical and feasible.	
Homework / Follow-up	Complete assigned perpendicular-bisector/Voronoi practice and write one real-world use of a Voronoi diagram. Bring 2–3 IA interests and one draft research question next lesson.		
Teacher's notes / Reminders	Students must bring a device, graphing calculator/GDC and notebook.		

Technology Integration	Microsoft Teams/OneNote; graph paper, rulers, coordinate grids, GDC/Desmos for plotting sites, constructing perpendicular bisectors and visualising Voronoi regions; IA brainstorming and planning template.
Cross-curricular links	Geography/Urban Planning: service catchments and nearest-facility regions. Computing: spatial algorithms and site allocation. Business/Logistics: delivery hubs and service coverage. Research skills: selecting a feasible IA question and considering data quality.
External Assessment Integration	International Baccalaureate: coordinate geometry, perpendicular bisectors and Voronoi diagrams as a spatial partitioning model; introductory preparation for the mathematical exploration / Internal Assessment.

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Wellbeing/ ATL Skills	**Thinking:** analyse distance, midpoint and gradient relationships; model and evaluate spatial decisions. **Communication:** justify constructions and explain reasoning. **Research:** brainstorm topics and assess data sources. **Self-management:** organise IA ideas and next steps. Social: collaborate and give peer feedback.
Standards	IB Mathematics: Topic 3 coordinate geometry, perpendicular bisectors and Voronoi diagrams; introduction to mathematical exploration skills for the Internal Assessment.
Inquiry/ TOK Question	How can mathematics help us decide which location is closest or best served? When we use a mathematical model to make a real-world decision, what assumptions and limitations should we consider?
Success Criteria (Measurable)	**Lesson 1: Perpendicular Bisectors** • **All students must** calculate the midpoint and gradient of a segment using given coordinates. • **Most students should** find the negative reciprocal gradient and determine the perpendicular bisector's equation. • **Some students could** justify the construction algebraically and verify the midpoint and perpendicular relationship. **Lesson 2: Introduction to Voronoi Diagrams** • **All students must** identify sites and describe a Voronoi region as points closest to one site. • **Most students should** construct a Voronoi diagram using perpendicular bisectors as boundaries. • **Some students could** explain why points on a boundary are equidistant from neighbouring sites and interpret vertices. **Lesson 3: Applications of Voronoi Diagrams** • **All students must** identify the nearest site for a point using a given diagram. • **Most students should** apply or construct a diagram to solve a contextual location problem and justify the result. • **Some students could** evaluate assumptions/limitations or predict how adding a site changes the regions. **Lesson 4: Introduction to Internal Assessment** • **All students must** identify the purpose of the Mathematics IA, recognise its main sections, and name the assessment criteria. • **Most students should** describe the purpose of each main section and briefly explain what the IA assessment criteria evaluate. • **Some students could** analyse a past IA to identify strengths and areas for improvement in its structure, mathematical communication, and alignment with the assessment criteria.
	Area: Citizenship
	Responsible and informed decision-making
	Implementation: Use Voronoi diagrams to consider fair and efficient access to public services such as clinics, schools or emergency facilities, while recognising that distance is only one factor in real-world access.